Lab 6 — Move, Copy, Rotate and Offset

WSQ Technical Drawing with AutoCAD (TGS-2021005538) · Topic 03: Modifying Objects

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Objective

LO2/LO3 — modify 2D drawings efficiently to build up a design (A3, K3)

Goal

Build practice drawing EX-4 by drawing one feature and then positioning the rest with MOVE, COPY (multiple), ROTATE with reference angle and OFFSET for the parallel profiles.

You'll produce

The completed EX-4 drawing produced with modify commands rather than re-drawing each feature.

Commands / tools: MOVE, COPY (Multiple/Array), ROTATE (Reference), OFFSET

Files in this lab

- ex4.dwg
- ex4-drawing-sheet.pdf
- ex4.png

Step-by-step

1. Open the EX-4 sheet (ex4-drawing-sheet.pdf) and spot the repeated features
2. Draw the base profile, then offset it to create the parallel inner/outer edges

   ```
   O → distance → pick side
   ```

3. Draw one hole feature and copy it to every dimensioned centre

   ```
   CO → base point → second points
   ```

4. Move a mis-placed feature precisely from its centre to the correct intersection

   ```
   M → base → destination
   ```

5. Rotate the angled feature about its base point, using Reference to match the sheet angle

```
RO → base → R → ref angle → new angle
```

6. Check against ex4.dwg and save your version

```
SAVEAS ex4-mine.dwg
```

Check your work

Feature spacings measure correct with DIST, offsets are at the specified distances, and the rotated feature sits at exactly the sheet's angle.

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